

Epidemiology

SI

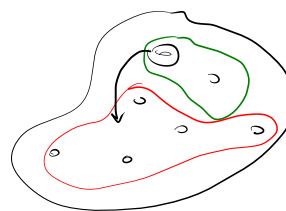
$S(t)$ = # suscettibili a t

$I(t)$ = # infetti a t

$N(t) = S(t) + I(t)$

$$\rightarrow s(t) = \frac{S(t)}{N}$$

$$\rightarrow i(t) = \frac{I(t)}{N}$$



- 1) Consideriamo dt molto piccolo. Affinché un S si infetti deve incontrare un I e (assumendo well mixed population)

prob. che un individuo sia infetto = $\frac{I}{N}$

- 2) Di conseguenza per un suscettibile il tasso di infezione è $\beta \frac{I}{N}$

- 3) # di nuovi infetti nel dt = $\beta S \frac{I}{N}$

$$\begin{cases} \frac{1}{N} \frac{dS}{dt} = -\beta \frac{SI}{N \cdot N} \\ \frac{1}{N} \frac{dI}{dt} = \beta \frac{SI}{N \cdot N} \end{cases}$$

$$\begin{cases} \frac{ds}{dt} = -\beta si \\ \frac{di}{dt} = \beta si \quad (*) \end{cases}$$

MA SAPPIAMO $s + i = 1$
 $s = 1 - i$

$N(t) = S(t) + I(t)$
 $\hookrightarrow \frac{dN}{dt} = 0 = \frac{dS}{dt} + \frac{dI}{dt}$
 $\hookrightarrow \frac{dS}{dt} = -\frac{dI}{dt}$

(*) $\frac{di}{dt} = +\beta i(1-i)$

$$\frac{di}{i(1-i)} = \beta dt$$

$$\left(\frac{A}{i} + \frac{B}{1-i} \right) di = \beta dt$$

$$\frac{A(1-i) + B \cdot i}{i(1-i)} = \frac{A + i(B-A)}{i(1-i)} \quad \begin{cases} A = 1 \\ B - A = 0 \end{cases} \quad \begin{cases} A = 1 \\ B = 1 \end{cases}$$

$$\int \left(\frac{1}{i} + \frac{1}{1-i} \right) di = \int \beta dt$$

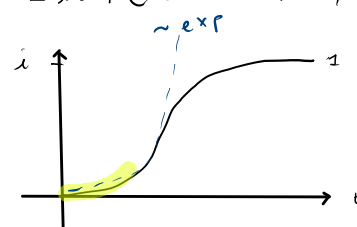
$$\ln(i) - \ln(1-i) = \beta t + \text{cost}$$

$$e^{\left(\ln \left(\frac{i}{1-i} \right) \right)} = e^{\beta t + \text{cost}}$$

$$\frac{i}{1-i} = c \cdot e^{\beta t} \quad i(t) = c e^{\beta t} - i(t) c e^{\beta t} \quad i(t) (1 + c e^{\beta t}) = c e^{\beta t}$$

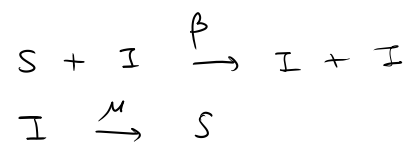
$$i(t) = \frac{c e^{\beta t}}{1 + c e^{\beta t}}$$

sigmoide



2 stati stabili $\rightarrow$ (banale) $i=0 \quad s=1$
 $i=1 \quad s=0$

SIS



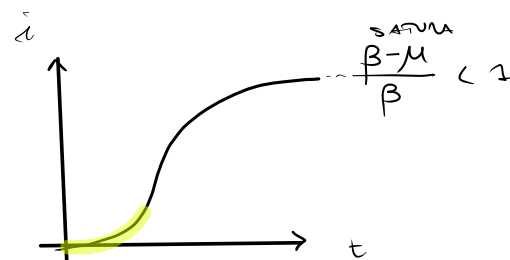
1 - IN AN INTERVAL dt # nuovi infetti = $\beta \frac{I}{N} S$

2 - " " # nuovi guariti = $\mu \frac{I}{N}$

$$\begin{cases} \frac{dS}{dt} = -\beta Si + \mu i \\ \frac{di}{dt} = \beta Si - \mu i \end{cases}$$

$$\rightarrow i(t) = i_0 \frac{(\beta - \mu) e^{(\beta - \mu)t}}{\beta - \mu + \beta i_0 e^{(\beta - \mu)t}}$$

SIGMOIDE



$$i + S = 1$$

3 - $t \approx 0$ EARLY PHASE $\rightarrow$ ALL SUSCEPTIBLE $S \sim 1$
 $i \ll 1$

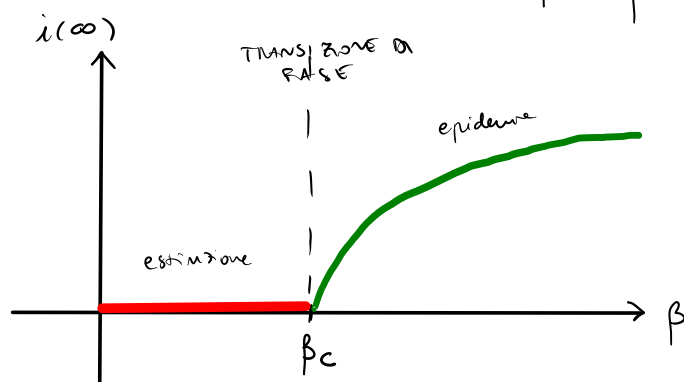
$$\frac{di}{dt} \sim (\beta - \mu)i \rightarrow i(t) \sim i_0 \cdot e^{(\beta - \mu)t}$$

$\left\{ \begin{array}{l} \beta < \mu \\ \beta > \mu \end{array} \right.$

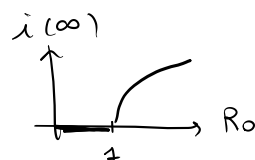
ESTINZIONE

EPIDEMIA

$$i(t) > 0 \Leftrightarrow \beta > \beta_c = \mu \quad \text{EPIDEMIC THRESHOLD}$$



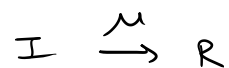
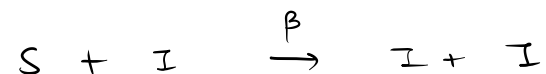
... OR $R_0 \equiv \frac{\beta}{\mu}$



SIR

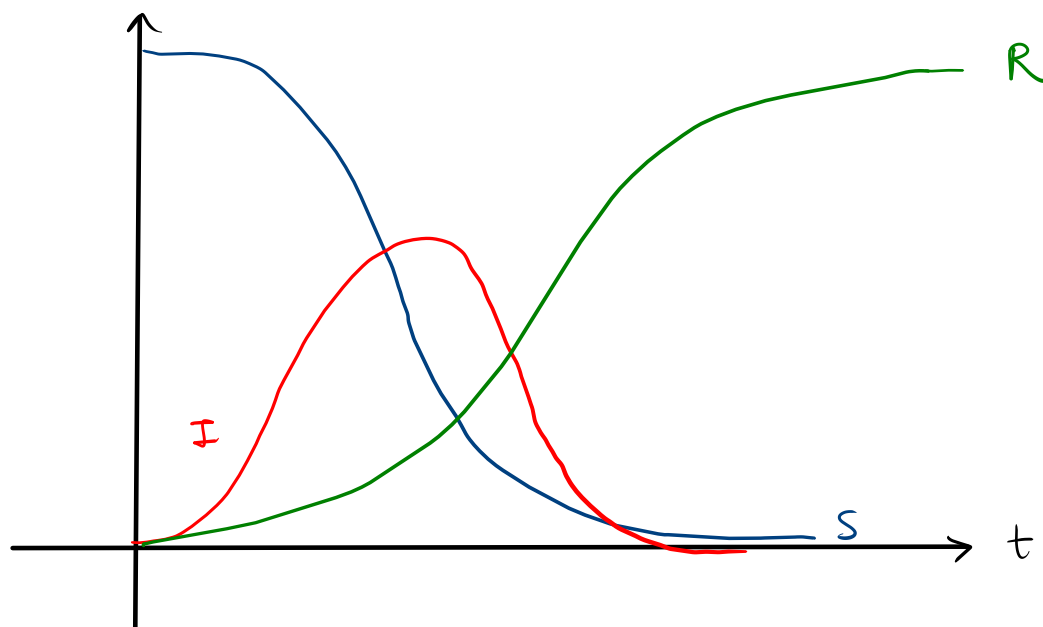


immunità permanente



$$\begin{cases} \frac{ds}{dt} = -\beta si \\ \frac{di}{dt} = \beta si - \mu i \\ \frac{dr}{dt} = \mu i \end{cases}$$

$$s + i + r = 1$$



$$1 - t \sim 0 \rightarrow s \sim 1$$

$$\frac{di}{dt} = \beta i - \mu i = i(\beta - \mu) \Rightarrow i(t) \sim i_0 e^{(\beta - \mu)t}$$

STESSO COMPORTAMENTO DI SIS
(β THRESHOLD ecc...)

$$2 - t \rightarrow \infty$$

$$\frac{ds}{dr} = -\frac{\beta}{\mu} s \Rightarrow s(t) = s_0 \cdot e^{-\beta/\mu \cdot r(t)}$$

$$t \rightarrow \infty \Rightarrow \begin{cases} i(\infty) = 0 \\ s(\infty) = 1 - r(\infty) = s_0 \cdot e^{-r(\infty)\beta/\mu} \end{cases}$$

Metodo di Eulero

per approssimare la soluzione di un ODE 1° ordine

$$\begin{cases} \frac{dy}{dt} = f(y(t), t) \\ y(t_0) = y_0 \end{cases} \rightarrow \frac{dy}{dt} \sim \frac{y(t_{n+1}) - y(t_n)}{\Delta t} = f(y(t_n), t_n)$$

$$y(t_{n+1}) = y(t_n) + \Delta t \cdot f(y(t_n), t_n)$$

È una formula approssimata!

ex

$$\frac{dy}{dt} = y$$

(sol esatta

$$\int \frac{dy}{y} = \int dt$$

$$\ln(y) = t + c \quad y = y_0 e^t = e^t)$$

$$y(0) = 1$$

$$\begin{matrix} t=0 \\ n=0 \end{matrix} \quad y_0 = 1$$

$$\begin{matrix} t=1 \\ n=1 \end{matrix} \quad y_1 = 1 + 1 \cdot 1 = 2$$

$$n=2 \quad y_2 = 2 + 1 \cdot 2 = 4$$

$$y_3 = 4 + 1 \cdot 4 = 8$$

$$y_4 = 8 + 1 \cdot 8 = 16$$

...



$$\text{sol esatta } e^4 = 54,6$$